

Ahmed Toba Mahmoud

Software Engineer — Backend Developer

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Professional Summary

Software Engineer with a strong foundation in computer science fundamentals, including data structures, algorithms, object-oriented programming, software design, and system architecture. Experienced in designing and developing scalable, secure, and production-ready software, with a specialization in Backend Development. Passionate about building reliable systems, solving complex engineering challenges, and continuously learning modern technologies.

Education

B.Sc. in Communications and Electronics Engineering

Beni Suef University | 2026 | Grade: Very Good

Graduation Project | Forging a Secure and Resilient Digital Backbone for a New Era of Academia

Grade: Excellent

[Github](#) | [Live Demo](#)

- Led a multidisciplinary engineering team in planning, architecting, and delivering a secure Smart University ecosystem spanning Backend Development, Cloud & DevOps, AI, IoT, Enterprise Networking, and Cybersecurity.
- Designed and implemented scalable backend services, RESTful APIs, real-time communication using Socket.IO, authentication and authorization mechanisms, and cloud-integrated system architecture.
- Developed a cloud-connected biometric attendance system using ESP32, Azure IoT Hub, secure device communication, offline synchronization, and real-time attendance processing. [Github](#)
- Engineered a multi-layer AI academic assistant featuring general academic Q&A, role-based live data retrieval, and a RAG-powered document intelligence system, with token budget optimization and defenses against prompt injection through secure role-based data access.

Professional Experience

Backend Developer Intern | Hi Deep AI | Beni Suef, Egypt

Sep 2025 – Dec 2025

- Collaborated directly with clients across the full Software Development Lifecycle (SDLC), contributing to business analysis, system analysis, solution architecture, implementation, testing, and project delivery.
- Designed, developed and enhanced scalable backend services, RESTful APIs, MongoDB data models, file upload functionality, and core application features within an Agile development team.
- Wrote unit tests, participated in code reviews, and resolved technical issues to improve software quality, reliability, and maintainability.

Projects

Natours — Full-Stack Tour Booking Platform

[GitHub](#) | *Node.js · Express.js · MongoDB · JWT · Stripe · Pug*

- Built a production-ready full-stack tour booking platform featuring secure RESTful APIs, server-side rendering (SSR), and responsive user experience.
- Implemented JWT authentication, Stripe payments, rate limiting, XSS/NoSQL protection, and other security best practices.

E-Commerce Platform — Full-Stack Shopping Platform

[GitHub](#) | *Angular · Node.js · Express.js · MongoDB*

- Developed a role-based full-stack e-commerce platform with product, cart, order, and authentication modules.
- Implemented secure backend APIs and integrated frontend workflows for a complete shopping experience.

Training & Certifications

- MEAN Stack Web Development (120 Hours) – National Telecommunication Institute (NTI) 2025
- Node.js, Express, MongoDB & More: The Complete Bootcamp – Udemy (Jonas Schmedtmann) 2025

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript, C++, C, SQL
- **Backend Development:** Node.js, Express.js, RESTful APIs, API Security, Socket.IO
- **Databases:** MongoDB, MySQL
- **Cloud & DevOps:** Microsoft Azure, Azure IoT Hub, Docker, AWS, Git, GitHub
- **AI Engineering:** RAG, Prompt Engineering, Prompt Injection Mitigation, Role-Based AI Systems, Token Budget Optimization, LLM Integration
- **IoT Development:** ESP32 Firmware Development, MQTT, Biometric Integration, IoT Device Communication
- **Core Fundamentals:** OOP, Data Structures & Algorithms, SOLID Principles, Software Design
- **Developer Tools:** Postman, Linux

Leadership & Volunteering

Co-Founder & Business Developer | Maymay Start-up Jan 2025 – Jan 2026

- Co-led business strategy, product development, and partnership initiatives to support startup growth.

Rally BSU | Beni Suef University

HR Manager Aug 2025 – Aug 2026
Led recruitment, team development, conflict resolution, and HR operations.

Marketing Vice Feb 2025 – Aug 2025
Directed marketing initiatives, campaign planning, and cross-functional team coordination.

Head of HR | GDSC EELU Feb 2024 – Aug 2024

- Managed recruitment, onboarding, mentoring, and student progress throughout Google Developer Student Clubs learning programs.

Languages

- Arabic:** Native
- English:** Professional Working Proficiency